Homework 7 (Optional) Answers

BSTA 550

Questions

Extra problems

4. (a) Poisson($\sum_{i=1}^{n} \lambda_i$) (b) Poisson(3λ) (c) $M_{3X}(t) = e\lambda(e^{3t} - 1)$ This is not an mgf of a common probability distribution.

(d) In (b) we are adding independent r.v.'s X_i, while in (c) we are adding dependent r.v.'s ($3X = X + X + X$; X is dependent with itself).